

Assignment 1: Literature Review

Agent-Based Models and Financial Vulnerability

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1 Introduction

The 2008 financial crisis showed that common risk tools such as Value-at-Risk (VaR) were not enough to measure systemic risk. VaR relies on historical data and assumes stable relationships. During crises, relationships between institutions change quickly and losses spread through feedback loops. Traditional stress tests improve on VaR, but they still rely on fixed scenarios and do not model how institutions react to each other over time.

Agent-based models (ABMs) offer a different approach. Instead of assuming equilibrium, ABMs simulate individual institutions and track how their actions affect the system step by step. This review examines how ABMs help explain fire sales, funding runs, leverage cycles, and network contagion.

2 Agent-Based Modeling of Financial Systems

Bookstaber, Paddrik, and Tivnan (2014, 2017) develop an agent-based model of financial vulnerability. Their framework includes cash providers, bank dealers, and leveraged investors. Each agent has a balance sheet and decision rules. When a shock occurs, agents adjust their positions, which changes prices and funding conditions. The model updates these effects over time.

A key result is that crisis severity depends more on how institutions react than on the size of the initial shock. Losses trigger balance sheet adjustments, which lead to further price declines. These feedback effects amplify stress.

The model also represents key financial functions such as maturity transformation, liquidity provision, leverage, and derivatives exposure. Even small networks can produce large spillovers when institutions hold overlapping assets or rely on similar funding sources.

3 Fire Sales and Contagion

Recent research studies how fire sales transmit stress across the system. Fire sales occur when institutions sell assets quickly to meet capital or liquidity constraints. Rapid selling pushes prices down, which weakens other institutions holding the same assets.

Caccioli et al. (2024) show that including non-bank institutions in contagion models increases estimated losses. Investment funds amplify price declines through asset sales, even when they are not directly connected through lending relationships.

Weber et al. (2020) combine fire sales and default contagion in one framework. They show that distress spreads not only through direct counterparty exposure but also through common asset holdings. This indirect contagion can cause losses even when direct links are limited.

Onaga et al. (2023) model fire sales using a continuous state framework. Instead of treating defaults as binary events, they track gradual declines in solvency. This allows more realistic simulations of how stress builds over time.

4 Leverage and Market Dynamics

Leverage plays a central role in amplifying crises. When asset prices fall, leveraged institutions must sell assets to maintain capital ratios. These sales lower prices further, creating a feedback loop.

Feinstein (2020) shows that capital regulation can increase instability during stress if many institutions try to deleverage at the same time. When constraints bind simultaneously, price impact becomes stronger.

Empirical evidence supports these mechanisms. High levels of margin debt and growth in leveraged products increase the risk of forced selling. However, during the COVID-19 shock, central bank interventions reduced the severity of fire sales by providing liquidity and absorbing selling pressure.

5 Network Structure and Policy

Modern financial systems are highly interconnected. Institutions are linked through loans, derivatives, funding markets, and common asset holdings. Network-based stress testing attempts to measure how shocks spread across these connections.

Recent regulatory stress tests incorporate cross-sector contagion. When banks, insurers, and funds are analyzed together, second-round effects increase total losses compared to

sector-by-sector analysis.

New research on multi-layer networks studies how different types of exposures interact. Risk can pass through several channels at once, such as derivatives combined with common asset holdings. These higher-order connections can create hidden vulnerabilities.

Policy research suggests that monetary policy and macroprudential tools work better when coordinated. Network-based indicators help identify when system-wide risk is building.

6 Research Gap and Hypothesis

Despite progress, many models assume similar behavior within each group of institutions. In practice, institutions differ in how quickly they respond to stress. Some rebalance portfolios immediately, while others adjust slowly. These differences may change how contagion unfolds.

Another limitation is limited use of high-frequency crisis data to test model predictions. Many models are calibrated to aggregate outcomes rather than to the actual sequence of losses during stress events.

This paper proposes the following hypothesis:

H1: Institutions that adjust their portfolios more quickly during stress events experience larger capital losses during fire sales, holding leverage and asset overlap constant. The effect is stronger when many institutions react at similar speeds, leading to synchronized asset sales and sharper price declines.

This hypothesis can be tested using high-frequency trading data and network measures of portfolio overlap. Regression or event study methods can estimate the relationship between reaction speed and realized losses. Agent-based simulations with heterogeneous reaction speeds can then compare simulated and observed outcomes.

7 Conclusion

Agent-based models provide a framework for studying systemic risk as a dynamic process driven by interactions among institutions. The literature shows that fire sales, leverage, and network structure amplify shocks through feedback loops. While current models capture many mechanisms, more work is needed to incorporate behavioral differences and to validate predictions with detailed crisis data.

Understanding how reaction speed and synchronization affect contagion can improve both risk management and macroprudential policy design.